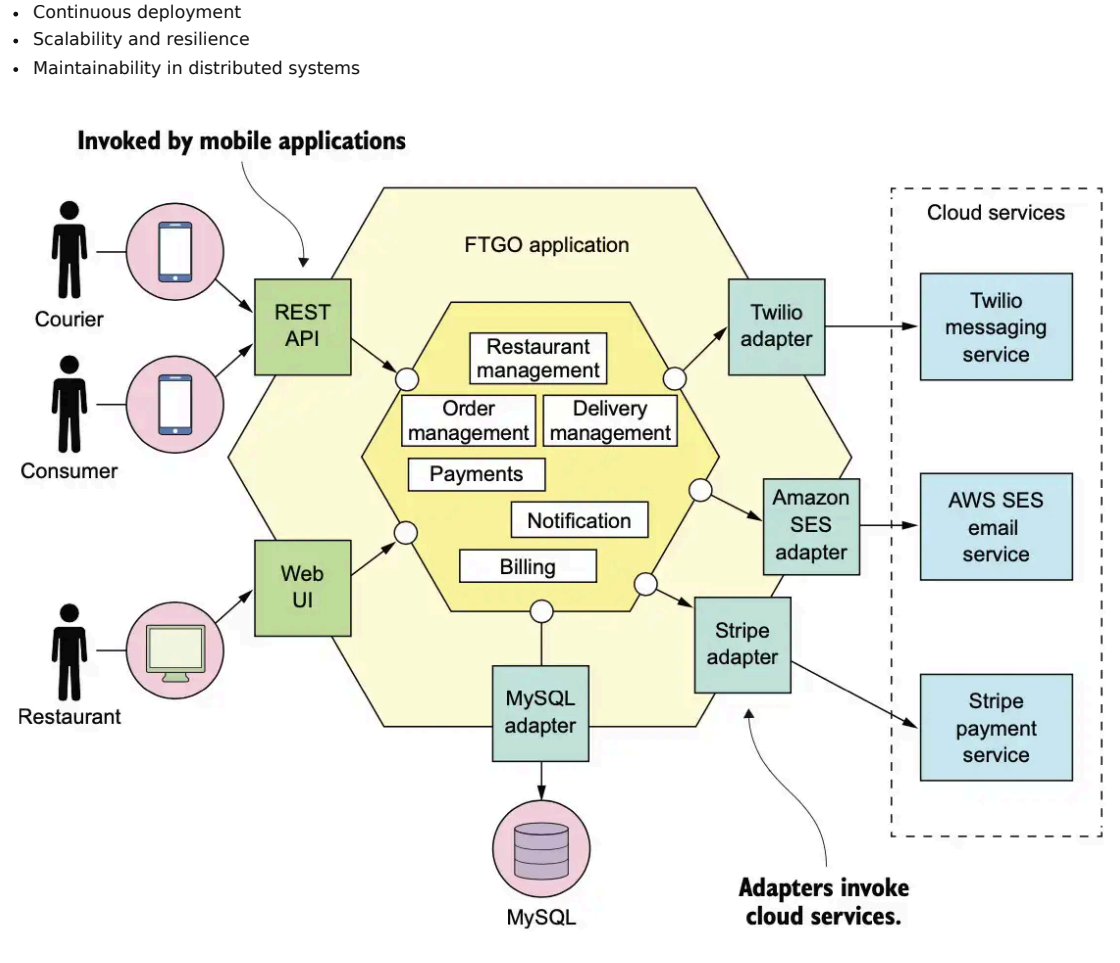
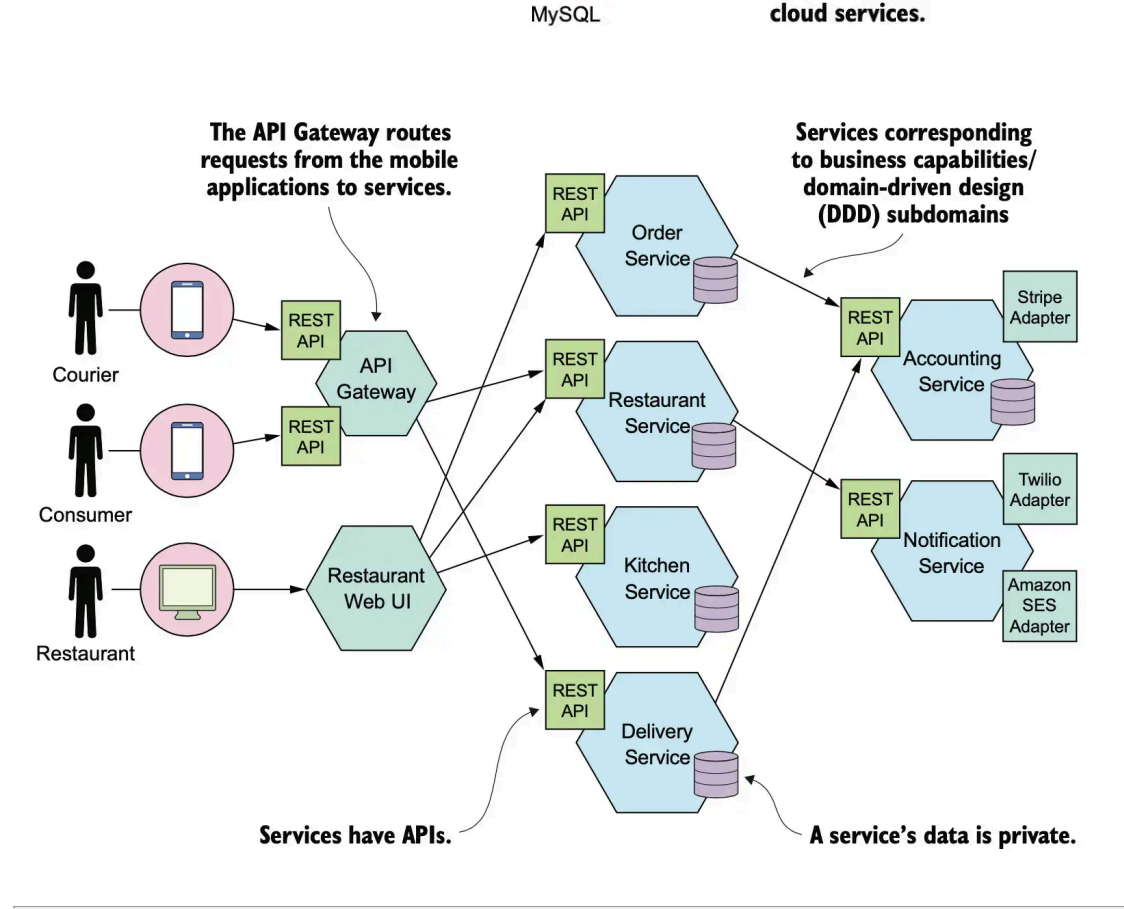
# The Twelve-Factor App

The Twelve-Factor App is a methodology for building **modern, scalable, cloud-native applications**.

- Portability across environments
- Continuous deployment
- Scalability and resilience
- Maintainability in distributed systems


Invoked by mobile applications
Courier
Consumer
Restaurant
REST API
Web UI
FTGO application
Restaurant management
Order management
Delivery management
Payments
Notification
Billing
MySQL adapter
MySQL
Twilio adapter
Amazon SES adapter
Stripe adapter
Cloud services
Twilio messaging service
AWS SES email service
Stripe payment service
Adapters invoke cloud services.


The API Gateway routes requests from the mobile applications to services.
Services corresponding to business capabilities/ domain-driven design (DDD) subdomains
Courier
Consumer
Restaurant
REST API
API Gateway
Restaurant Web UI
Order Service
Restaurant Service
Kitchen Service
Delivery Service
Accounting Service
Stripe Adapter
Notification Service
Twilio Adapter
Amazon SES Adapter
Services have APIs.
A service's data is private.

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## I. Codebase

- One codebase tracked in version control
- Many deploys from the same codebase

**Benefits:**

- Traceability and collaboration
- Simplified CI/CD
- Consistent releases across environments

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## II. Dependencies

- Explicitly declare and isolate dependencies

**Key Ideas:**

- Use dependency managers (e.g., Maven, Gradle, pip)
- Avoid relying on system-level packages
- Ensures reproducibility and portability

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## III. Config

- Store configuration in the environment

**Examples:**

- Database URLs
- API keys
- Credentials

**Benefits:**

- Separation of code and configuration
- Easier deployment across environments

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## IV. Backing Services

- Treat external services as attached resources (*An attached resource is something the application does not own or embed internally, but connects to at runtime through configuration.*)

**Examples:**

- Databases
- Message brokers
- Caches

**Key Principle:**

- Easily replace services without code changes

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## V. Build, Release, Run

- Strict separation of stages

**Stages:**

1. Build → compile and package
2. Release → combine build with config
3. Run → execute the application

**Benefits:**

- Repeatable deployments
- Reduced risk of errors

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## VI. Processes

- Execute the app as one or more stateless processes

**Key Concepts:**

- No shared memory
- State stored in external services

**Advantages:**

- Scalability
- Fault tolerance

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## VII. Port Binding

- Export services via port binding

**Implication:**

- Application is self-contained
- No external web server required

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## VIII. Concurrency

- Scale out via process model

**Example:**

- Multiple instances behind a load balancer

**Benefits:**

- Elastic scalability
- Improved resilience

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## IX. Disposability

- Fast startup and graceful shutdown

**Goals:**

- Quick scaling
- Resilience to failures

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## X. Dev/Prod Parity

- Keep development, staging, and production similar

**Focus Areas:**

- Time
- Personnel
- Tools

**Benefits:**

- Reduced deployment issues

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## XI. Logs

- Treat logs as event streams

**Practices:**

- Do not manage log files
- Send logs to centralized systems

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## XII. Admin Processes

- Admin tasks must:

- Use the same codebase

- Use the same configuration

- Run in the same environment

**Examples:**

- Database migrations
- Maintenance scripts

**Benefits:**

- Consistency with production environment

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## Resources

- https://www.12factor.net/
- https://www.youtube.com/watch?v=FryJt0Tbt9Q